

Structured Knowledge versus Parameters

Gemma 4 E2B with a curated knowledge graph vs. Gemma 4 12B on questions about recent multilingual NLP research

1 Hypothesis

Currently the development of LLMs seems to lie in making models bigger, at growing cost and still without grounding in checkable facts: Bender et al. asked whether language models can be *too big* [1]. The alternative is to give a model external knowledge [2]. Knowledge graphs (KGs) store such knowledge as compact (head, relation, tail) facts whose quality can be measured [3], and they are widely studied as a remedy for knowledge gaps and hallucinations [4, 5, 6].

Research question

Can a small model with a small, curated KG (**Gemma 4 E2B + KG**, 2.3B effective parameters) answer factual questions as well as a larger model without it (**Gemma 4 12B**)?

Hypothesis

With the knowledge graph, the performance of the small model (2.3B) is comparable to the larger one's (12B); the KG narrows the gap.

Contribution: a curated, reproducible KG and question set, built from papers on multilingual NLP published before and after the models' knowledge cutoff.

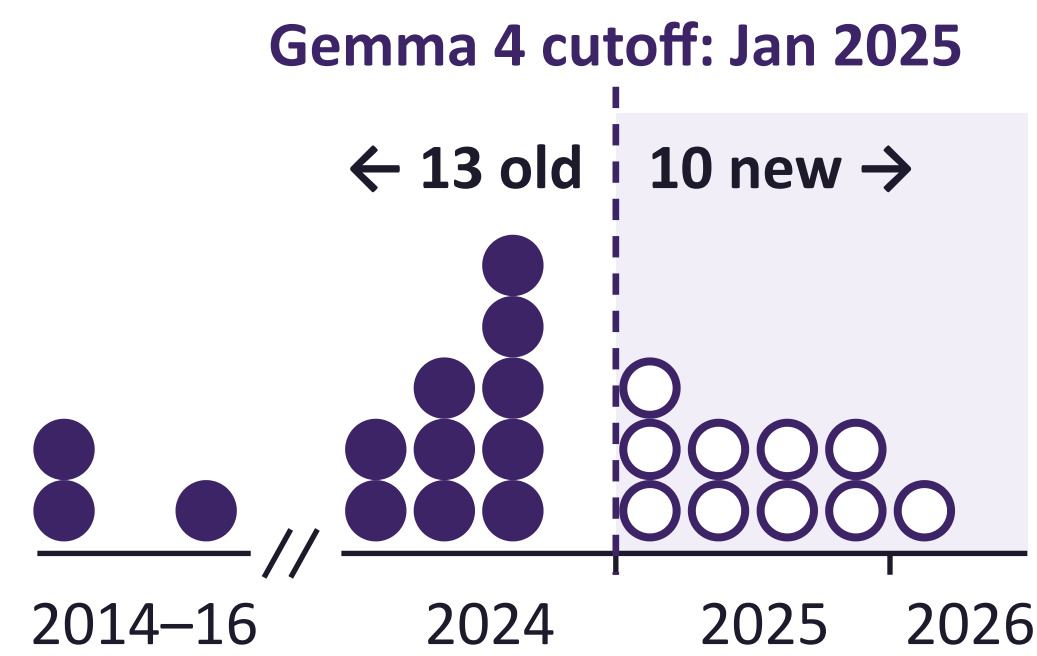


Fig. 1: Corpus papers by first public date. Papers after the cutoff cannot be known to the models.

2 Methodology

1 Corpus

23 papers → 116 passages

23 papers on multilingual NLP, mostly from *TACL* and *Computational Linguistics*. We use the abstract, introduction and experimental-setup section of each (116 passages, ~36k words). **Old/new:** first public version before/after January 2025 → 13 old, 10 new papers.

2 Triple extraction

576 candidates

Qwen3.5-9B turns each passage into (head, relation, tail) triples with 8 relations (e.g. *covers_language*, *evaluated_on*, *has_statistic*). *Why Qwen:* a different model family from the tested Gemma models, so their knowledge cannot leak into the KG. Candidates are kept only if head and tail appear in the passage.

3 Curation

338 triples = KG

KG quality has to be measured, not assumed [3]: 576 candidates were checked against the passages. 54% were usable as extracted, 59% after rewording → **338 unique triples**.

4 Questions

221 → 182 questions

221 multiple-choice questions drafted **from the passages, not from the KG**, so the KG is not guaranteed to hold the answer. After checking: **182 questions** (108 old, 74 new).

5 Evaluation

E2B E2B + KG 12B

Gemma 4 E2B and 12B (*why:* one family, two sizes, known cutoff) via Ollama, temperature 0, all conditions in one GPU session. **+KG:** all triples whose entity is named in the question are given as “Facts”. Accuracy (chance 25%); non-answers count as wrong.

Reproduce: github.com/viczon7/nlp-kg-comparison

Q: What is the parameter count for all models in the Goldfish suite?

A) 500M ✓ B) 1B C) 32B D) 7B

E2B	“I do not have specific, real-time knowledge about ... a proprietary suite like ‘Goldfish’.”	✗
E2B + KG	Facts: Goldfish suite has_statistic models have 500M parameters → A	✓
12B	D	✗

Fig. 2: One question, three conditions. One of the two retrieved facts holds the answer and lets the small model answer correctly.

3 Results

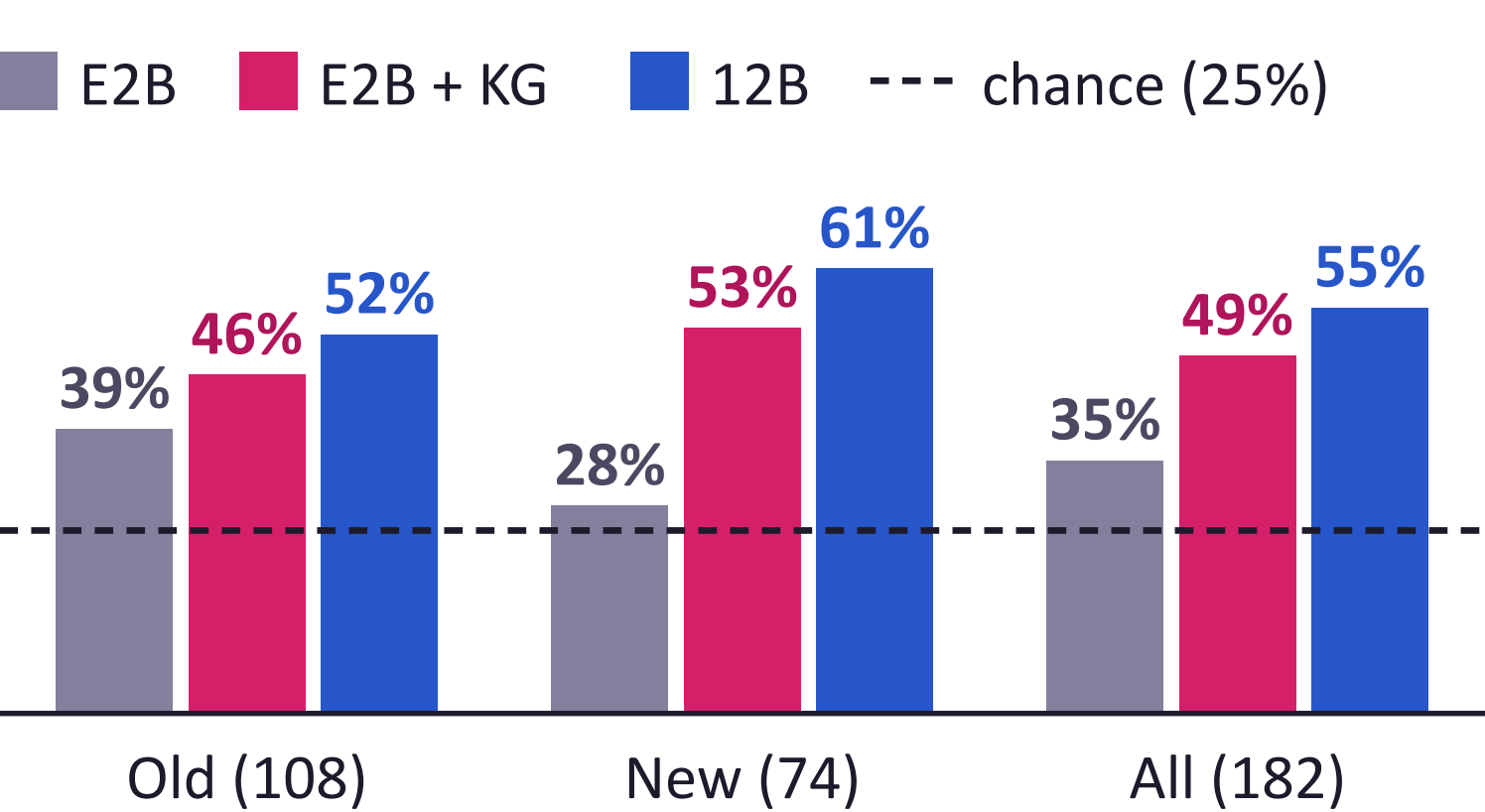


Fig. 3: Accuracy on 182 questions. The KG lifts the small model from 35% to 49%.

Table 1: Accuracy on all 182 questions, and the share of replies without an answer letter.

	Accuracy	No answer
E2B	35%	36%
E2B + KG	49%	29%
12B	55%	0.5%

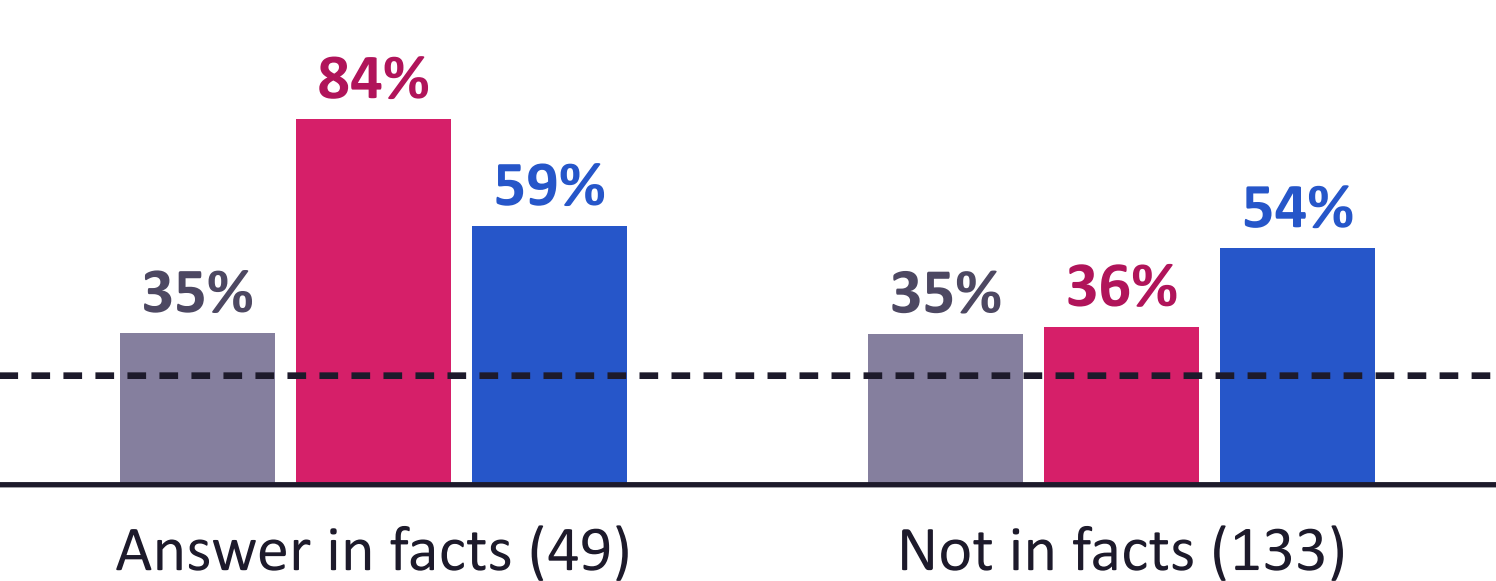


Fig. 4: When the KG holds the answer, the small model beats the larger one (84% vs. 59%).

RQ: The KG raises E2B from 35% to 49%. Where the KG holds the answer, E2B + KG beats 12B (84% vs. 59%); elsewhere 12B is better (54% vs. 36%). On old papers, the KG shrinks the gap to 12B from 13 to 6 points.

Hypothesis: the KG narrows the gap to 12B from 20 to 6 points.

Unexpected: E2B gave no answer letter 36% of the time, mostly by refusing. 12B scored higher on new papers (61%) than on old ones (52%), although it cannot know them: many questions seem answerable from the options alone.

4 Discussion & Conclusion

Limitations

Knowledge graph

- Only 27% of questions have their answer in the retrieved facts (old 15%, new 45%).
- Built from abstracts, introductions and setups only. Raw extraction precision was 54%, and multi-hop reasoning was not tested.
- Facts are found by exact name matching, which misses paraphrases and facts stored under other names.

Questions

- Only 182 multiple-choice questions.

Evaluation

- Non-answers (refusals, or replies cut off at 30 tokens) count as wrong. E2B often gives no answer, 12B almost never, so part of the gap measures willingness to guess.
- Only one domain and one model family.

Conclusion

For factual questions about recent research, a small model grounded in 338 curated facts came close to a 5× larger one, and beat it whenever the facts covered the question. Coverage, not model size, is the bottleneck: by extrapolation, ~40% coverage would match the larger model. Structured knowledge can stand in for parameters, but only where it covers the question.

Future work

- Extract from full papers and test on a fresh question set.
- Represent entities with KG embeddings [7, 8], so facts are found by similarity instead of exact names, and combine KG and text evidence [9].
- Add multi-hop questions over a graph database.

Take-home

A 338-fact knowledge graph brought a 2.3B model to within 6 points of a 12B model (49% vs. 55%), and made it better whenever the graph held the answer (84% vs. 59%).

[1]–[9]: references in the appendix.